PLUGGED IN

GETTING THE OLD BRUSH OFF

Toothbrushes: Lower prices and dentists' acceptance are helping power a shift in use from manual to electric.

By MICHELLE BETTON
SUN STAFF

More Americans are buying electric toothbrushes than ever before thanks to less-expensive technology, increased marketing and growing approval from dentists.

Sales grew 23 percent between 2002 and last year, according to Information Resources Inc., a Chicago-based retail-tracking group.

"If you buy the nice quality ones, it's great," said Savvy Palmer of Baltimore, an electric toothbrush user who believes the technology cleans better.

Consumers also bought fewer manual toothbrushes, with sales falling 6 percent between 2002 and last year, Information Resources reported.

Even so, about 180 million manual toothbrushes were sold last year — nearly six times as many as their electric cousins. Marketing experts consider cost a major reason.

"[Manual] toothbrushes are fine," said Ruth Louie of Baltimore, who said electric brushes are too expensive for her taste. "I replace them every few months and move on."

The total value of both markets is almost $800 million a year, but the growth in sales clearly is on the technological end of the spectrum, according to Information Resources.

A study last year showed little difference between results achieved with the two brushing methods. But as costs for electric toothbrushes have become more affordable, consumers seem more willing to embrace them. And dentists are increasingly recommending them to patients.

LLOYD FOX : SUN STAFF

Crest's SpinPro line of battery-powered electric toothbrushes, priced under $20, includes a Spider-Man model for youngsters.

Dr. Khin Phillips, a Baltimore dentist, has been recommending that her patients use electric toothbrushes for about three years. She uses a Braun Oral-B power toothbrush herself.

Phillips said electric toothbrushes are better at plaque removal and help patients get in the habit of brushing one tooth at a time. Consumers also are more willing to spend between $70 and $140 on the technology if their dentist recommends it, marketing experts and dentists said.

The advent of battery-powered toothbrushes, which cost between $6 and $20, has also fueled the growth in sales, letting consumers in on the power toothbrush trend at a fraction of the price of high-end models.

Electric toothbrushes have been around since the 1940s, starting with a device called the Toothmaster, said Dr. Scott Swank, curator for the National Museum of Dentistry in Baltimore. The first successfully marketed electric toothbrushes were unveiled in 1961. The Squibb Broxodent was a brush that needed to be plugged in, while General Electric produced a cordless rechargeable model.

By 1995, up to a quarter of all toothbrush sales were electric, Swank said. Sonic and ultrasonic technology emerged during the early 1990s and between 2000 and 2001, battery-operated toothbrushes emerged.

Toothbrush companies have worked to promote the new technology during the past several years through advertising campaigns. Marketing efforts also target young children with themed battery-operated brushes, sporting characters such as Spider-Man and Hello Kitty.

Area dentists said two of the most popular power toothbrushes are the Philips Sonicare line and the Braun Oral-B series.

The Sonicare brush ($119 to $139) uses sonic technology and side-to-side motion to move its bristles at high speeds. The Oral-B Professional Care 8000 ($120) uses a rotating oscillating head with high-speed pulsation to clean teeth.

The makers say that their cutting-edge brushes reduce plaque and gingivitis more than any manual brush. But results of a January 2003 study by the American Dental Association casts some doubt on such claims.

The ADA study, conducted to learn whether power toothbrushes clean more effectively than their manual counterparts, tested various brushing techniques as well as products. The study found that only the brush with the rotating oscillating head — the Braun Oral-B — showed significant benefits over manual brushes in reducing plaque and gingivitis. But the study found that those benefits were modest.

But Dr. Clifford Whall, director of the ADA's Seal of Acceptance Program, said that parts of the study were not "robust" enough. He added that a more definitive study is needed because it is not clear how much better one brush is than another.

Some dentists say they recommend electric toothbrushes because their motion is gentler than manual brushing and lessens irritation of gums. Some also said those with timers help people who don't brush for the full two minutes the ADA recommends.

Whall said that people may like power brushes better because they are easier to use. Manual brushes should be held at a 45-degree angle to clean teeth properly, he said, but a power brush does not have similar requirements.

And it's good oral care that's the bottom line, no matter what kind of brush it is, said Leslee Williams, ADA manager of media services.

"The key is that the user has to use the brush properly," Williams said.

GAMEROOM

ALGERINA PERNA : SUN STAFF PHOTOS

While football is serious business to them, testing "Madden NFL 2005" — with teammate Ray Lewis on the cover — is all in fun for Baltimore Ravens Randy Hymes (left), Terry Jones and Ron Johnson.

'NFL 2005' gets tryout with Ravens

By RASHEIM T. FREEMAN
SPECIAL TO THE SUN

Several Baltimore Ravens players tried out the new video game *Madden NFL 2005* this week to test what is billed as a loosely based version of the team's own defense.

Electronic Arts Sports, the game's California-based publisher, holds "Madden Day" every year at the training camp of the player featured on the cover. All-Pro linebacker Ray Lewis is the cover boy for this year's edition, which officially goes on sale today.

With nearly 37 million copies sold, the "Madden" franchise is the No. 1-selling series of football video games in North America. And the new version seems to be eagerly anticipated: According to gaming site IGN.com's GamerMetrics, an interactive tool that tracks interest in games, the *Madden NFL 2005* Web page was the most-viewed sports title during the past month.

EA Sports executives said the new game includes better graphics, a Defensive Playmaker Control feature and the ability to force turnovers by using the Hit Stick to make hard tackles and force fumbles.

The enhancements were popular with players such as rookie wide receiver Clarence Moore.

Ravens rookie Clarence Moore gives the video game's souped-up defense high marks.

"The overall defense is better," he said. "You just can't run around and cut back as fast on offense like you used to do in the old version."

Starting defensive back Will Demps said he uses the technology to "get a feel for how the other NFL teams play."

Backup quarterback Josh Harris said elements "like the play-action stuff and some of the fade routes," reminded him of the Ravens' style of play.

Former coach and player John Madden and EA Sports started the video game franchise in 1989 to teach users how to play football in a three-dimensional video environment. Madden said he wanted to combine a teaching tool with the recreation of video.

Since 2001, the Madden NFL franchise has featured athletes on the cover of the video game, some of whom have suffered injuries. That has led to talk of a "cover jinx" for players who appear on the game.

Last year's cover athlete, Michael Vick of the Atlanta Falcons, broke his shin bone during a preseason game against the Ravens the week the game was released. Others players who have graced the game's covers have suffered serious injuries or had subpar seasons.

But Lewis said he did not believe in any jinx.

"I only play my game one way, which is full speed," Lewis said. "If somebody is going to get hurt it's just not going to be me."

Madden NFL 2005

Price: $50

Rating: E for Everyone

Platforms: Six platforms, including Sony PlayStation 2, Microsoft Xbox and Nintendo GameCube.

MIKE HIMOWITZ

Routers, firewalls good hack-attack armor

[*Himowitz, from Page 1D*]

we discuss firewalls and routers, it's a good idea to know a little bit about how the Internet works — but don't worry, we'll keep geek-speak to a minimum.

Every computer connected to the Internet has a unique numeric address theoretically visible to every other computer. This Internet Protocol (IP) address is usually expressed as a series of four numbers separated by periods, such as 68.128.31.5.

This addressing system is the key to the power of the Internet — it's the reason why it's as easy to connect to a computer in Tasmania as it is to one across town.

Internet service providers assign IP addresses to their customers. Home users typically have IP addresses assigned on the fly whenever they dial in to an ISP such as America Online or Earthlink, or when they first connect their PCs to broadband cable or DSL modems.

Because the Internet was designed in a more innocent age as an open system for sharing both information and computing resources, the Internet Protocol makes it easy for machines to communicate.

When one computer wants to talk to another, it sends a message to the receiving machine's IP address, and the other machine responds.

Typically, computers use different communications channels, known as ports, for different purposes, such as Web browsing, e-mail, file transfer, or instant messaging. The problem, in this age of malicious hackers, is that without any type of security, outsiders can take advantage of these channels and flaws in the computer's operating system to poke around in your files, steal information, plant mischievous programs, or even take control of your PC. This is particularly true if you have file or printer sharing enabled on a local network.

How does an intruder find you? Typically with software that scans IP addresses looking for a basic response (known as a "ping") and then probing for "open" ports that respond to commands or requests for information. Computers running Microsoft Windows have proved particularly vulnerable to these attacks — note the monthly security patches that Microsoft issues to plug these holes. But other operating systems have vulnerabilities, too.

Most vulnerable are PCs connected to the Internet directly through a cable or DSL modem, particularly if they're left running 24 hours a day.

That's where routers and firewalls come in. They work together to minimize the danger, and their features often overlap.

A router is designed to connect one network to another. The typical home router is a small box with four to six connections for network cables and often a wireless antenna that allows computers with wireless adapters to join the network.

As its name implies, the router passes Internet information packets along the proper path to their destination, like a train or truck dispatcher. But routers designed for the home and small offices also allow multiple PCs to share an Internet connection. Typically, you'll connect one or more computers to the router and plug the router into your cable or DSL modem.

Once installed, a router serves as a "front man" for all the computers on the local network behind it, even if the "network" consists of a single machine. The router contacts the ISP, obtains a single IP address for the whole network and passes information to the proper PC on the inside. It also allows computers on the home network to share information without going through the Internet.

Although they can be configured to block or pass different types of data packets, most routers are set up by default to hide the ports that hackers access. The router allows outside packets through only from communications sessions your computer initiates — such as sending e-mail, browsing the Web, or starting an Internet chat program.

This is also one of the basic functions of a firewall. In real life, a firewall is a fireproof barrier that separates one structure or room from another to keep a blaze from spreading. In the virtual world, it's a piece of hardware or software that provides additional levels of protection from outsiders. Some routers have firewalls built in to protect an entire network, but typical home firewalls are built through software that protects a single PC.

In addition to hiding a PC from outsiders, a firewall can keep unauthorized programs from accessing the Internet. These include programs planted by intruders or e-mail viruses that can turn your computer into a spam "zombie," launch denial-of-service attacks on other computers or transmit personal information to a rogue Web site.

When a program it doesn't recognize tries to access the Internet, a firewall will alert you and ask whether you want to allow or deny it. Most firewalls also create logs of attempts to contact your computer, allowing you to see the addresses of potential break-in artists.

If you have a broadband connection, you'll need a router to connect more than one PC to the Internet. But even with a single computer, a router is a good idea. They're easy to set up, and basic wired models are available for as little as $30. Wireless routers cost anywhere from $50 to $200, depending on their wireless connection speed and other features. Among the major manufacturers are Linksys, D-Link and Netgear.

Firewall software is part of multipurpose Internet security suites from Norton, McAfee and other publishers. These usually include an anti-virus program, spam blocker and spyware filter, too.

You also can purchase a standalone firewall program. The best-known is ZoneAlarm, largely because the publisher offers a basic but very effective version free of charge (www.zonelabs.com). Microsoft Windows XP comes with an elementary firewall, but it's not much good because it doesn't block outbound traffic.

It's unfortunate that we need so many levels of protection online these days. But it's the price we pay for defending our castles from the online Visigoths.